

DATA SCIENCE – FINAL PRACTICAL TASKS

Data Analysis, Pandas, Visualization, SQL and Power BI

Data Science Instructor: Ananda Rimal

Final Submission Deadline: October 6, 2026

Important:

All four tasks must be completed and submitted before the final deadline.

Final Deadline: October 6, 2026

Students should organize all files properly inside one Google Drive folder and submit the Drive link.

Task 1 – Handwritten Mathematics Notes

Write all the **Mathematics topics discussed in class** completely by hand in your copy/notebook.

- Write all Mathematics notes discussed in class.
- Include all important formulas.
- Include the examples solved during class.
- Show numerical calculations properly and step by step.
- Use proper headings for each topic.
- Keep the notes clean, complete and well organized.

Important:

This task must be completed **only as handwritten notes**.

No Jupyter Notebook or digital notebook is required for Task 1.

Task 2 – IPL Data Analysis: 200 Questions

Use the provided **IPL datasets** and solve a total of **200 data analysis questions**.

All 200 questions must be solved inside **one Jupyter Notebook**.

Part A – Pandas Analysis: 150 Questions

Solve **150 questions** using Pandas.

Part B – Data Visualization: 50 Questions

Solve another **50 questions** using data visualization.

Use:

- Matplotlib
- Seaborn

Total for Task 2:

Section	Questions
Pandas Data Analysis	150
Matplotlib + Seaborn Visualization	50
Total	200

File Name:

Task_2_IPL_Pandas_Visualization.ipynb

Task 3 – IPL SQL Analysis: 50 Questions

Import both IPL datasets into your SQL database as **two separate SQL tables**.

For example:

- IPL Matches Table
- IPL Deliveries Table

Create appropriate relationships between the tables wherever applicable.

Solve **50 SQL questions** using the SQL concepts discussed in class.

Put the screenshots and make a pdf and upload in drive

Task 4 – IPL Power BI Dashboard

Use the same **two IPL datasets / SQL tables** used in Task 3.

Connect the SQL database with **Power BI** and import both tables.

Create the required relationship between the tables and build a complete IPL Data Analysis dashboard.

The Power BI report must contain exactly **5 report pages**.

Submission Checklist

Before submitting, make sure that:

Task 1 complete Mathematics notes are handwritten.

All 150 Pandas questions are completed.

All 50 Matplotlib/Seaborn questions are completed.

Total 200 IPL Python questions are completed.

Both IPL datasets are imported into SQL.

Primary Key and Foreign Key are properly defined.

All 50 SQL questions are completed.

SQL screenshots are included.

SQL is connected with Power BI.

Both IPL tables are available in Power BI.

Power BI contains all 5 report pages.

The Power BI .pbix file is submitted.

All files are organized inside one Google Drive folder.

Google Drive access permission is enabled.

FINAL DEADLINE: OCTOBER 6, 2026

Complete all tasks before the deadline.

**Data Science Instructor
Ananda Rimal**